

TERAHOP

Open CPX MSA for AI Scaling

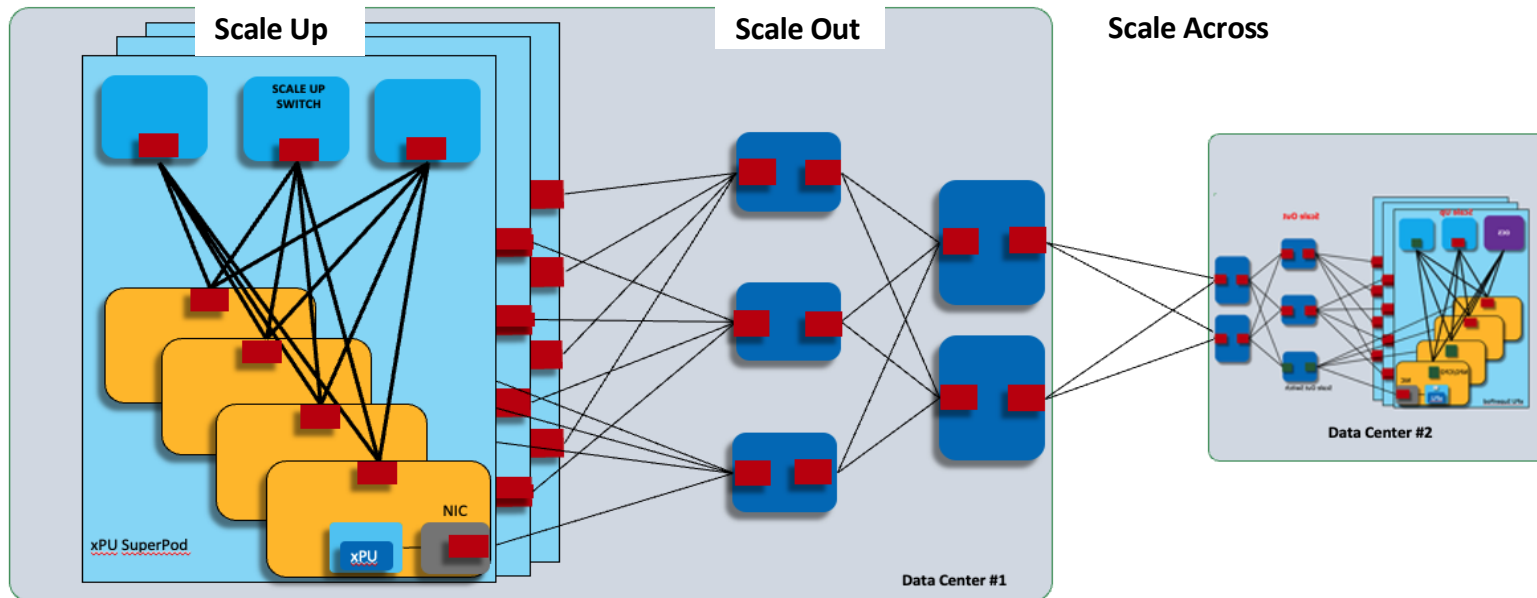
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Hot Interconnects
August 20, 2026

AI Scaling Challenges

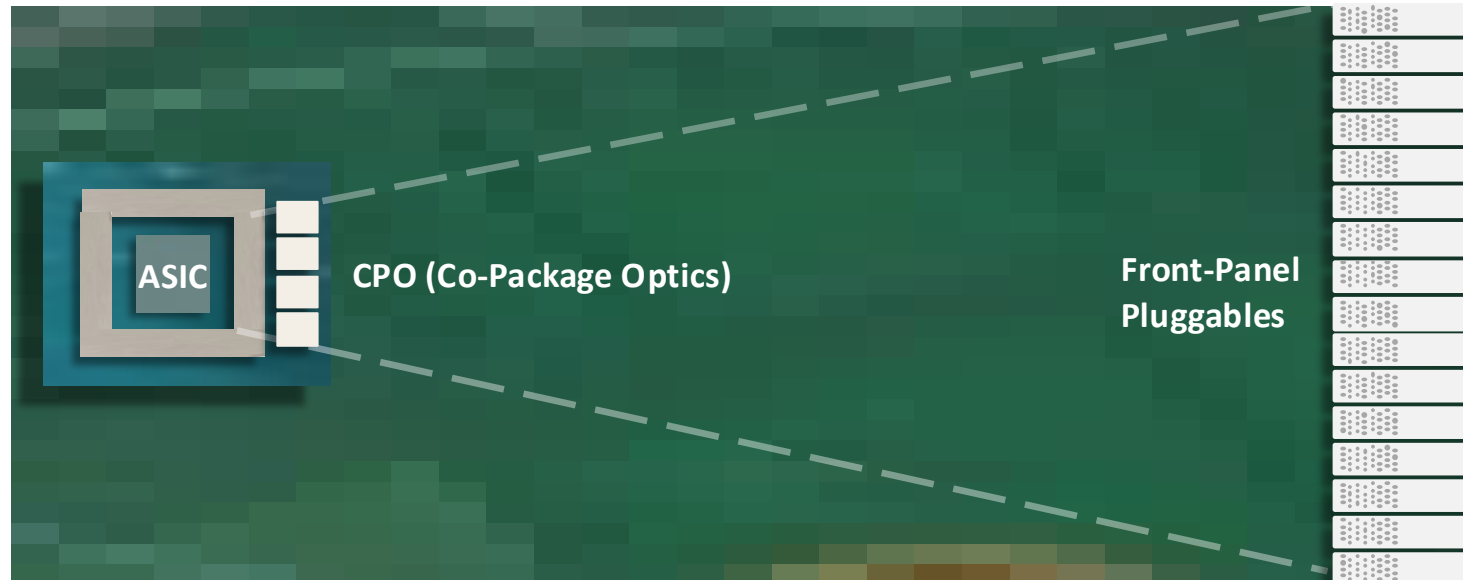
- Optical connectivity solutions required to support AI scaling from Scale Up, Scale out and Scale Across
- AI scale up transitioning from copper to optics with opportunity and challenges
 - Hi capacity ($\sim 10\text{Tbps/xPU}$)
 - Hi density (limited escape, space and system face plate...)
 - Lower power (Pj/b)
 - High reliability/availability and supply resiliency
 - Fast “time to volume”: no time for slow ramp..



Optical Connectivity Form Factor Evolutions and Challenges

Moving optical transceivers closer to ASIC: From Front panel pluggable to CPO (Co-Packaged Optics)

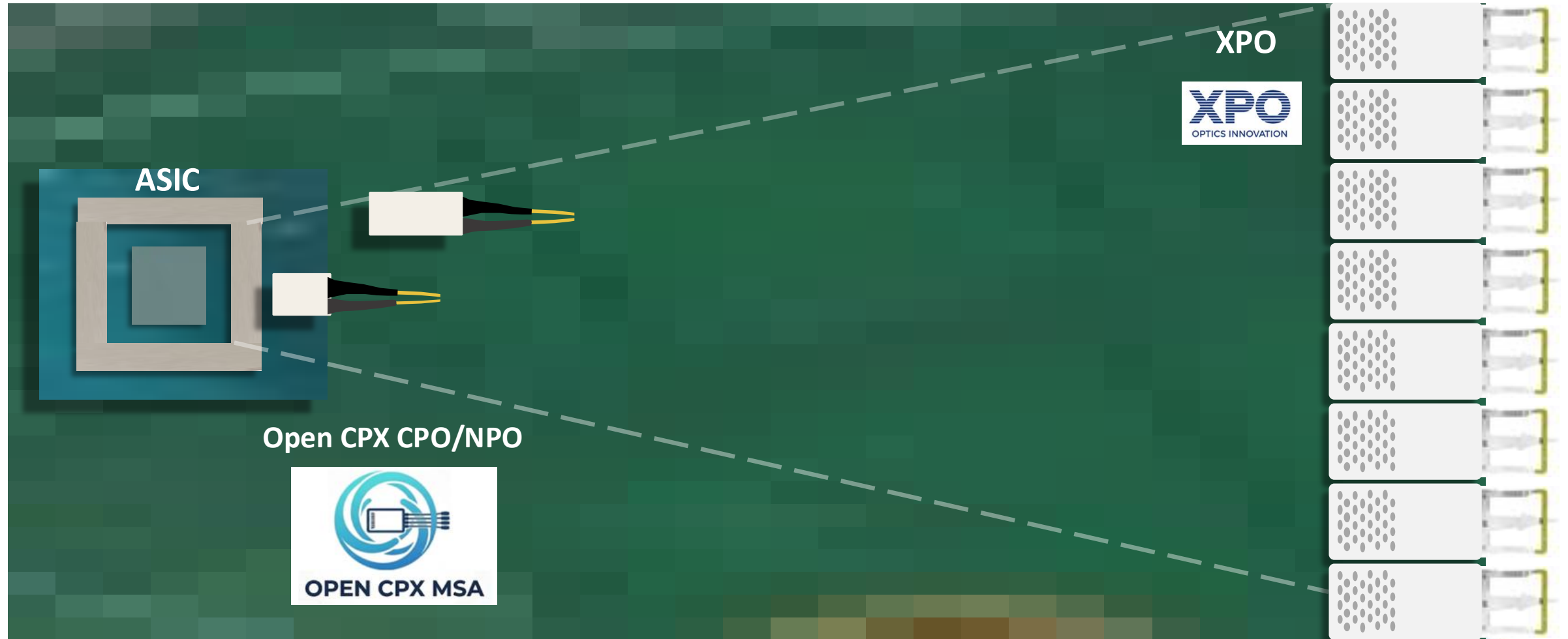
- Pros: lower copper channel loss → lower power power dissipation
- Cons/Challenges
 - Proprietary Single vendor lock-in
 - Tight integration with ASIC
 - Compounding yield challenge
 - Lacking technology diversity (e.g., narrow options of MMR modulator with external laser)
 - Poor serviceability
 - Do not allow mix of copper and optics
 - Lacking mass production deployment track record
 - Lacking supply resiliency



Open Optical Connectivity Solutions for AI

Open Optical Solutions addressing AI Scaling Challenges:

- XPO MSA: New High capacity, high density front panel pluggable with integrated liquid cooling
- Open CPX MSA: New open socket CPO/NPO for economical, large-scale deployments



Open CPX MSA: Open CPO/NPO for AI Scaling



The Open CPX MSA founding members include Ciena, Coherent, Marvell, Molex, Samtec, and TeraHop (alphabet order).



The Open CPX MSA early Contributor members include Accton Technology, Intel, NextHop Systems and TE Connectivity Solutions. (alphabet order)

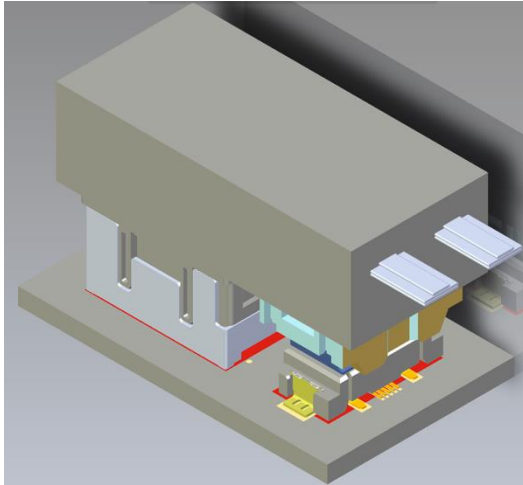


Incumbent CPO	Open CPX MSA
Proprietary, Single vendor lock-in	Multi-sourced with 80+ companies joining
Tight integration with ASIC	Enabling De-coupling with ASIC, supporting NPO, CPO, IOA
Compounding yield challenges	Individually tested before installation
Lacking Tech. diversity	Tech. options: - Mod: MZM, MMR; LD source: ILM vs. ELM; OE tech: Sipro vs. VCSEL, TFLN, etc.
Poor Serviceability	Pluggable socket allow economical repairing/replacement
Do not allow mix of Copper and Optics	Defined sockets supporting both optics, and copper
Lacking mass production deployment	Mature, mass deployed silicon photonics and field proven lasers
Supply resiliency	- MSA founders and members represent large share of optical industry - Diverse ecosystem with optical components, IC suppliers, and system integrators

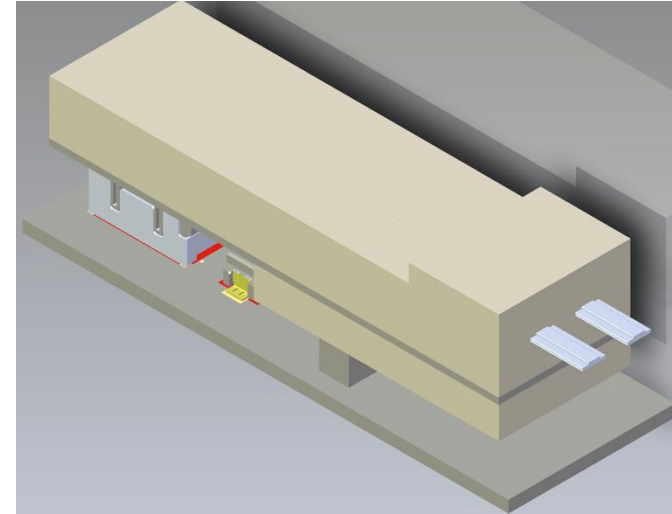


Open CPX Form Factors

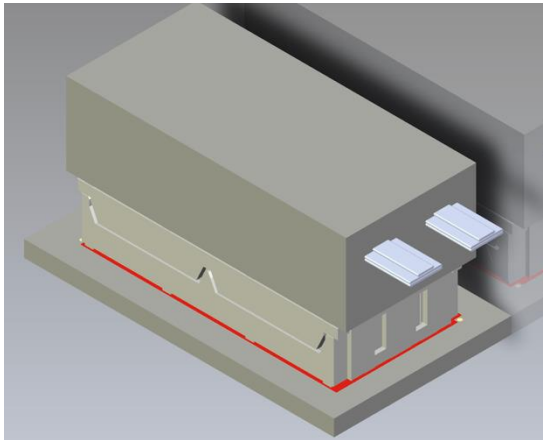
Type 1 Connector, ELM



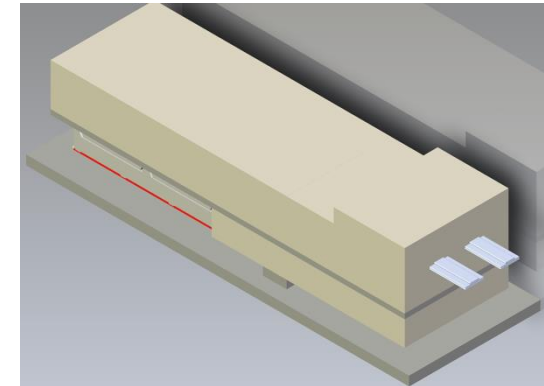
Type 1 Connector, ILM



Type 2 Connector, ELM

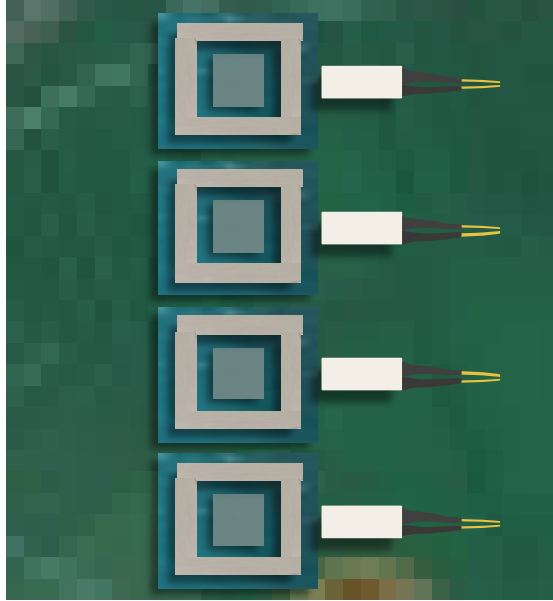


Type 2 Connector, ILM

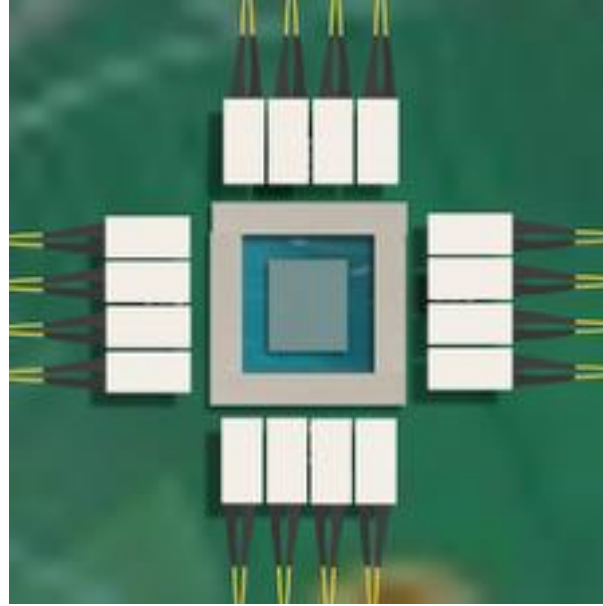


Open CPX Supporting Broad NPO, CPO and IOA Use Cases

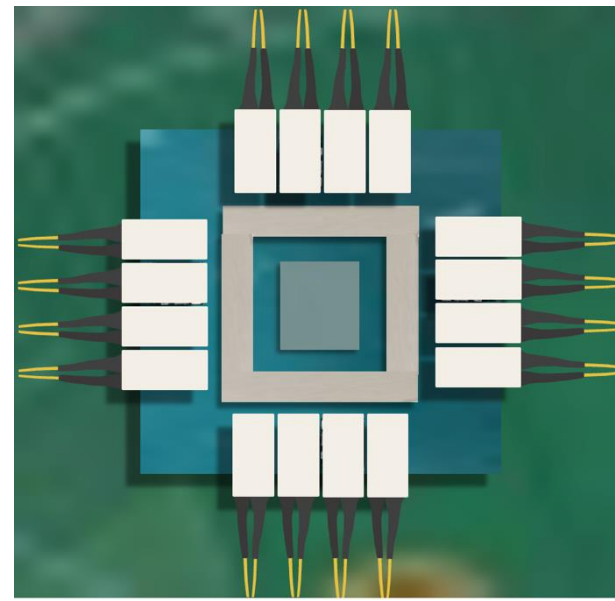
NPO for xPU Scale up



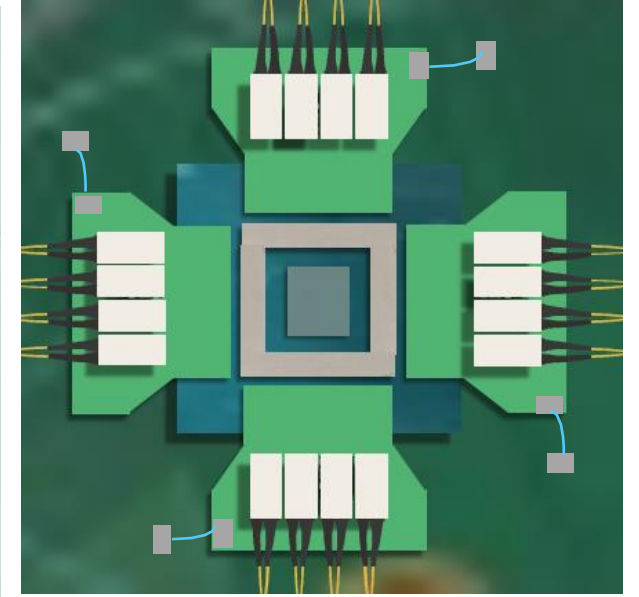
NPO for Switch Scale up/Out



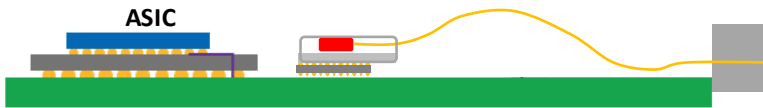
CPO for Switch Scale up/Out



IOA (Integrated Optical Attachment)

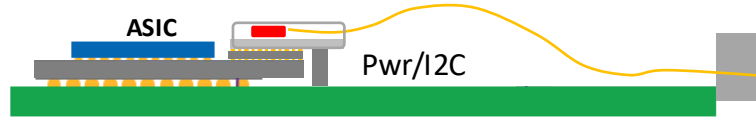


Open CPX NPO



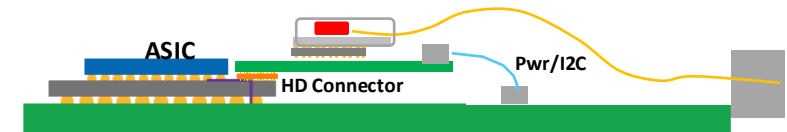
- Optical Engine on Host PCB

Open CPX CPO



- Optical Engine on ASIC substrate

Open CPX IOA

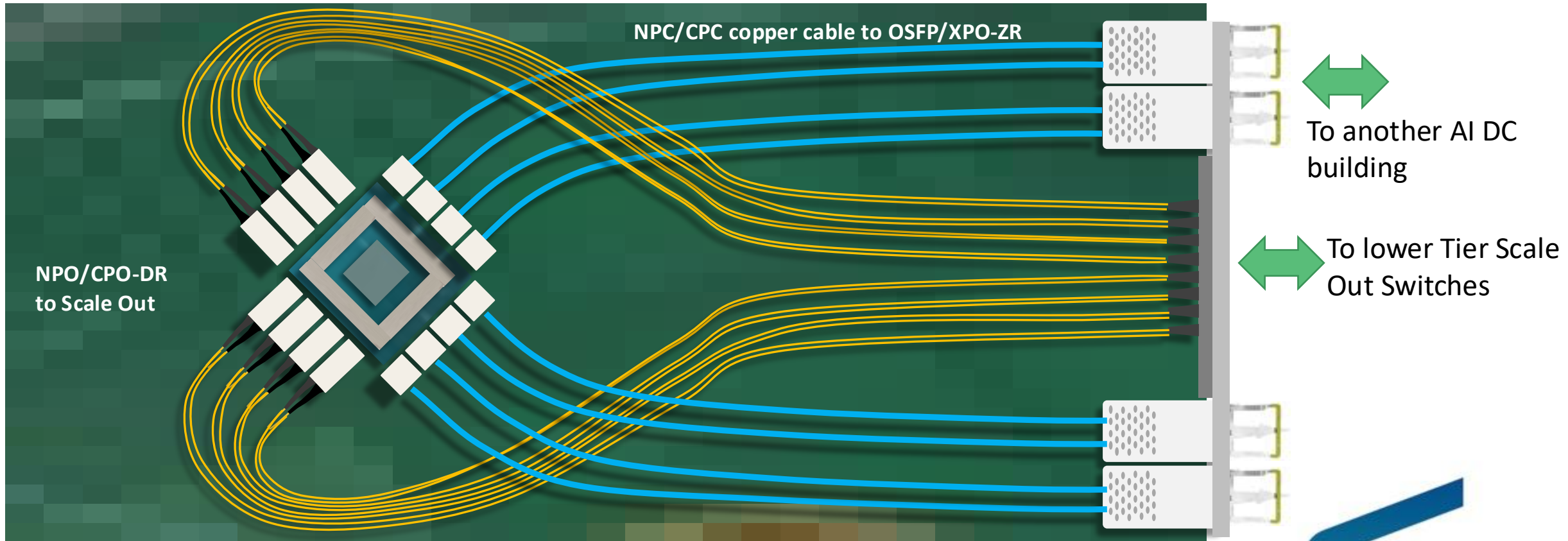


- Optical engine on HDI board attach to ASIC substrate via high density, high speed low loss connector



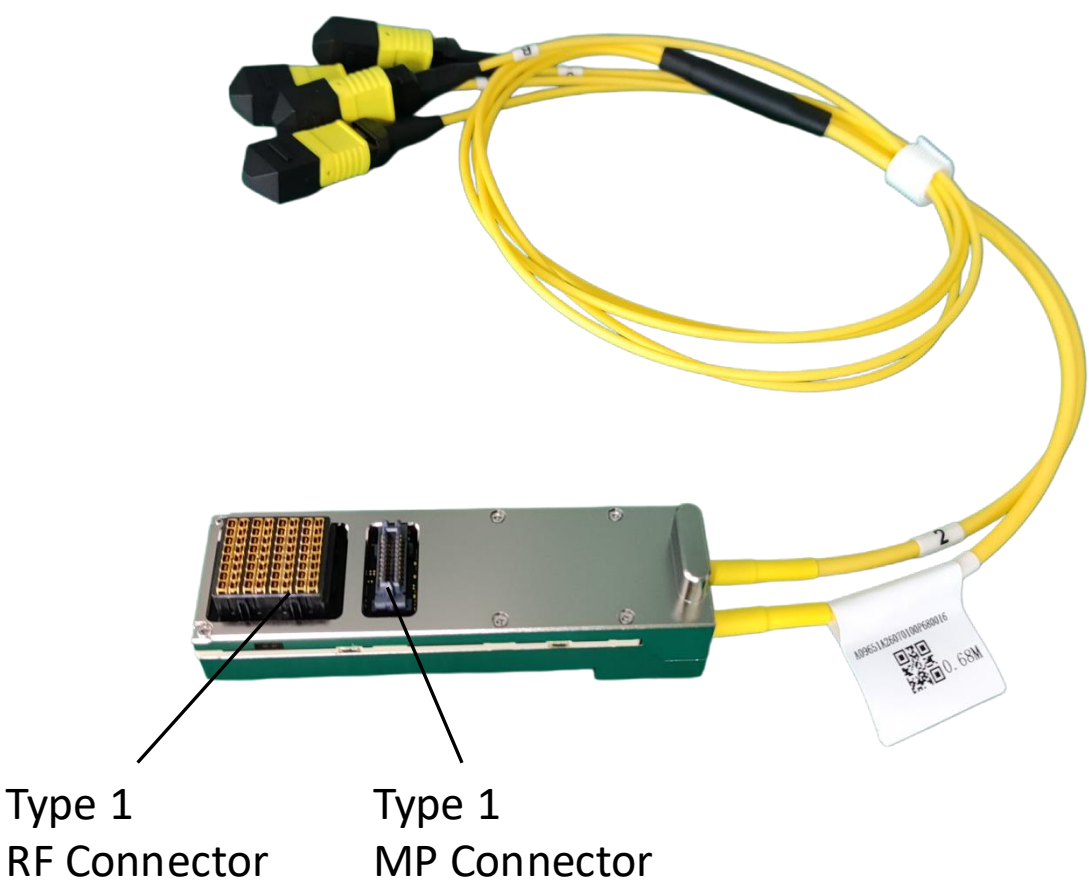
Open CPX Enabling Mix of Copper/Optics

- Mix of Copper and optics required for AI scaling in the next 5 years
 - “Use copper when you can, use optics when you must”
- Enabling different size of scale up domain: Copper in rack, optics cross rack
- Enabling Optical mix use cases: e.g., NPO-DR to scale out ; OSFP/XPO-ZR to scale across

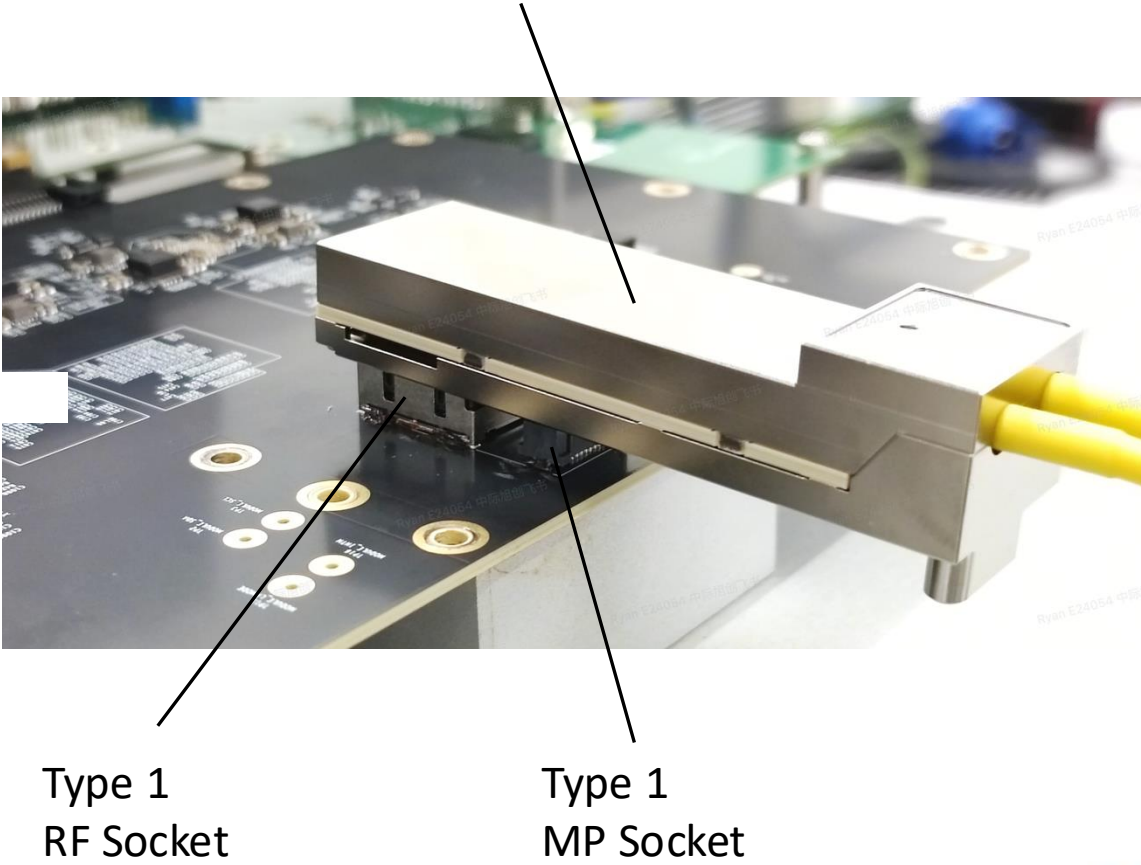


Example of Open CPX 6.4T-NPO

TeraHop “Diablo-1” 6.4T-8DR8 Open CPX NPO

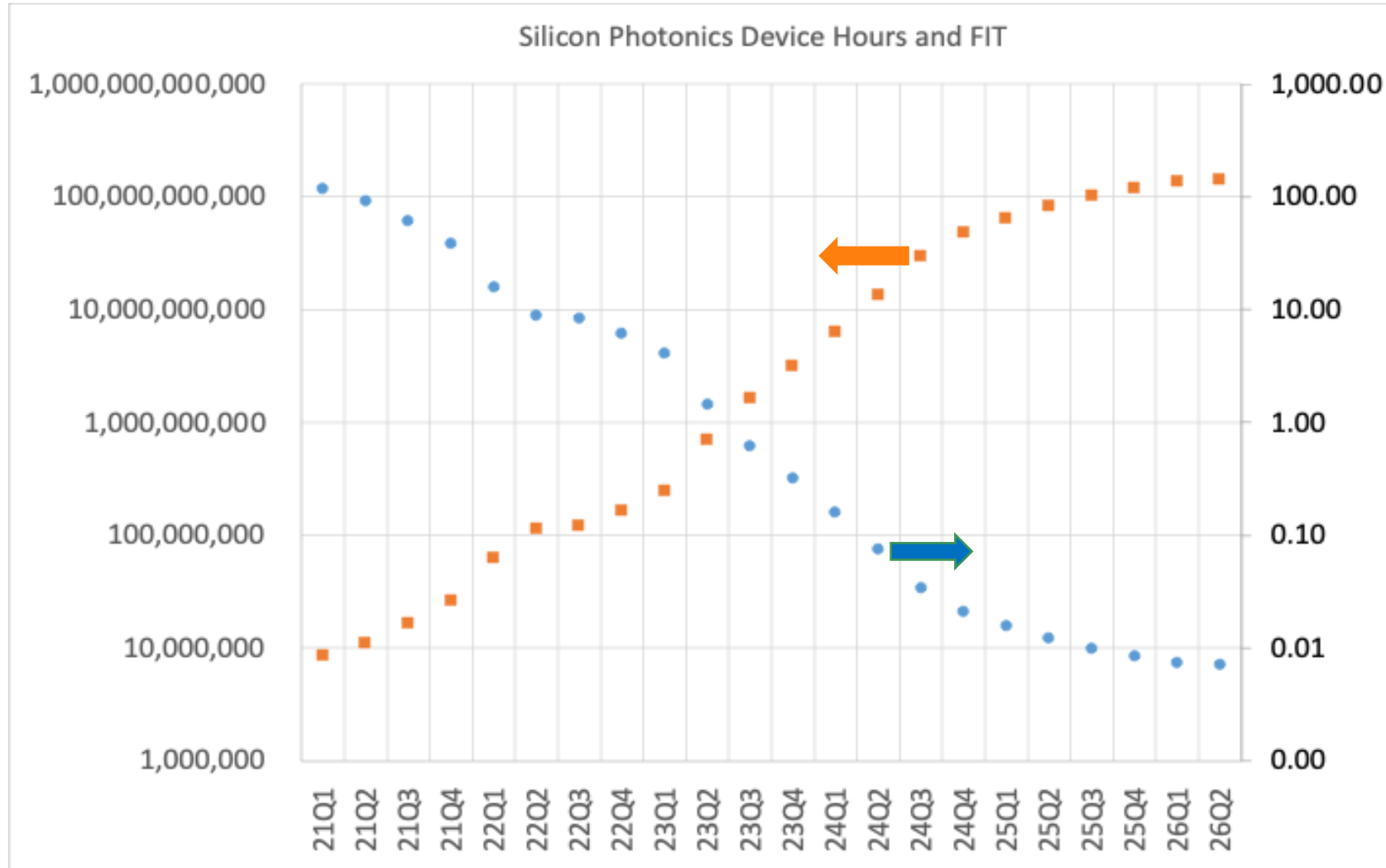


Silicon Photonics Optical Engine with ILM



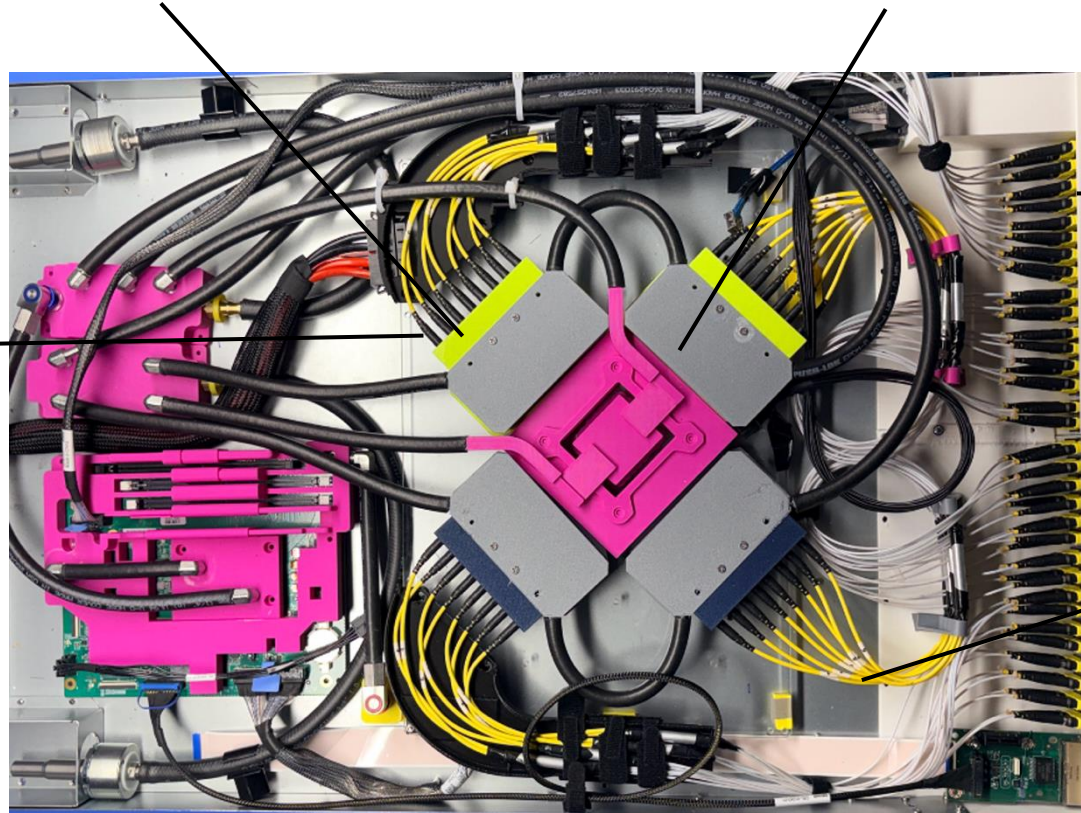
Silicon Photonics Supporting Open CPX Optical Engine

- TeraHop shipped > 25M Silicon Photonics optical transceivers, with PIC >100B Device-Hours, FIT< 0.01
- Not all “silicon photonics” created equal: maturity and high rel. needs field data



Industry 1st 100T Switch With Open CPX NPO

- NPO with integrated laser make system integration simple, clean, high yield, reliable
 - NPO with integrated lasers with low FIT
 - Late-stage assembly of fully tested NPOs
 - Dedicated Liquid-Cooling Cold-plate for NPOs



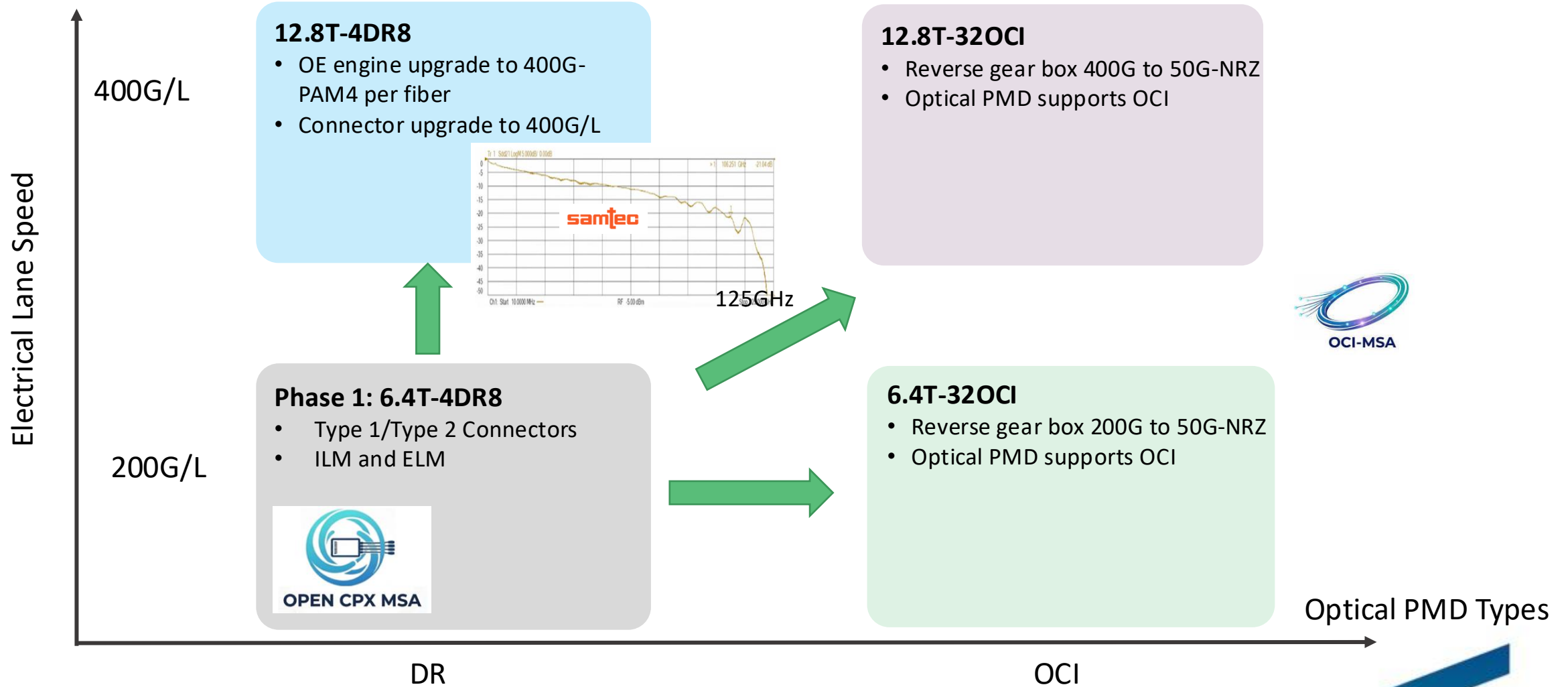
- Integrated fiber pigtail, eliminating failure-prone optical interface

- Clean Front Panel with optical I/Os
 - No ELSFPs taking up large front panel space
- Clean SMF fiber bundles to the front panel
 - No extra PM fiber from ELSFPs

Industry 1st NPO based 100T switch Design-Concept by Arista, with Open CPX 6.4T NPO from TeraHop



Open CPX MSA Potential Roadmap



Summary

- Open CPX MSA created to enable open CPO/NPO for AI scaling
 - Addressing many deployment pain points for large scale AI deployment

Multi-sourced with 80+ companies joining

Enabling De-coupling with ASIC, supporting NPO, CPO, IOA

Individually tested before installation

Tech. options: - Mod: MZM, MMR; LD source: ILM vs. ELM; OE tech: Sipro vs. VCSEL, TFLN, etc.

Pluggable socket allow economical repairing/replacement

Defined sockets supporting both optics, and copper

Mature, mass deployed silicon photonics and field proven lasers

- MSA founders and members represent large share of optical industry
- Diverse ecosystem with optical components, IC suppliers, and system integrators

- Welcome industry-wide participations: www.OpenCPXMSA.org



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THANK YOU

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